



Module 1 challenge

1st Attempt:

1. Fill in the blank: A data professional includes _____ visualizations in a presentation because stakeholders want the visualizations to reflect the latest data as it becomes available.

1 / 1 point

- ☐ static
- ☒ dynamic
- ☐ published
- ☐ embedded

✓ Correct

2. You are a junior data analyst presenting quarterly earnings to shareholders, including people who have visual and hearing impairments. What steps should you take to help your visualizations be accessible to everyone? Select all that apply.

1 / 1 point

- ☒ Label data directly whenever possible

✓ Correct

- ☐ Minimize contrast between colors
- ☒ Provide text alternatives

✓ Correct

- ☒ Reduce the amount of information in the presentation

✓ Correct

3. A data analyst in marketing visualizes the distribution of income among different customer segments. The graphic represents how often the data values fall into certain ranges. What type of visualization have they created?

1 / 1 point

- ☐ Pie chart
- ☒ Histogram
- ☐ Venn diagram
- ☐ Bar graph

✓ Correct

4. A data professional working in online sales finds that products frequently discussed on social media tend to sell more units. They discover this because two variables in their data rise and fall at the same time. What is this an example of?

1 point

- ☐ Visualization
- ☒ Causation
- ☐ Correlation
- ☐ Tabulation

✗ Incorrect

Review the video [about powerful data visualizations](#). ↗

5. Which of the following statements accurately describe key elements of data visualizations? Select all that apply.

1 / 1 point

- ☒ Lines can be used to help build the structure for a visualization.

✓ Correct

- ☒ Shapes in visualizations should always be two-dimensional.

✓ Correct

- ☐ Lines in visualizations must be straight.

- ☒ The hue of a color is basically its name, such as green or blue.

✓ Correct

6. While creating a graphic, a data analyst chooses a bright color for a large category and a more muted color for a smaller category. Which element of effective visualization does this describe?

1 / 1 point

- ☒ Sophisticated use of contrast
- ☐ Clear meaning
- ☐ Subtitles
- ☐ Refined execution

✓ Correct

7. Fill in the blank: Donut charts use _____ to combine the individual parts in a visualization and display them together as a whole.

1 / 1 point

- ☐ patterns
- ☐ color shading
- ☐ sizing
- ☒ data composition

✓ Correct

8. Which of the following statements correctly describe bar graphs? Select all that apply.

1 / 1 point

- ☒ The vertical line of a bar graph usually placed to the left is called the y-axis.

✓ Correct

- ☐ Bar graphs use segments to represent the proportions of each data category compared to the whole.
- ☒ The horizontal line of a bar graph, usually placed at bottom, is called the x-axis.

✓ Correct

- ☒ Bar graphs use size contrast to compare two or more values.

✓ Correct

2nd Attempt:

1. Fill in the blank: A data professional includes _____ visualizations in a presentation because stakeholders do not want the visualizations to change unless they choose to edit them.

1 / 1 point

- ☐ interactive
- ☐ dynamic
- ☐ linked
- ☒ static

✓ Correct

2. You are a junior data analyst at a government agency presenting a new policy to the public. What steps should you take to help your visualizations be accessible to all people, including those with visual impairments? Select all that apply

1 / 1 point

- ☒ Reduce the amount of information in the presentation

✓ Correct

- ☒ Label data directly whenever possible

✓ Correct

- ☒ Provide text alternatives

✓ Correct

- ☐ Minimize contrast between colors

3. A data analyst in marketing visualizes the distribution of income among different customer segments. The graphic represents how often the data values fall into certain ranges. What type of visualization have they created?

1 / 1 point

- ☐ Bar graph
- ☒ Histogram
- ☐ Pie chart
- ☐ Venn diagram

✓ Correct

4. A data professional working for a manufacturing company finds that products with certain temperature variations are more likely to have defects. They discover this because two variables in their data rise and fall at the same time. What is this an example of?

1 / 1 point

- ☐ Tabulation
- ☐ Visualization
- ☒ Correlation
- ☐ Causation

✔ Correct

5. Which of the following statements accurately describe key elements of data visualizations? Select all that apply.

1 / 1 point

- ☒ Colors can be described by their hue, intensity and value.

✔ Correct

- ☒ Value indicates how much light is being reflected.

✔ Correct

- ☒ Lines can be used to add visual form to the data.

✔ Correct

- ☐ Shapes in visualizations should always be three-dimensional.

6. While creating a graphic, a data analyst chooses a bright color for a large category and a more muted color for a smaller category. Which element of effective visualization does this describe?

1 / 1 point

- ☐ Refined execution
- ☐ Clear meaning
- ☐ Subtitles
- ☒ Sophisticated use of contrast

✔ Correct

7. Fill in the blank: Pie charts use _____ to combine the individual parts in a visualization and display them together as a whole, which can help reveal patterns and trends not visible in the original datasets.

1 / 1 point

- ☒ data composition
- ☐ color shading
- ☐ patterns
- ☐ sizing

✓ Correct

8. Which of the following statements correctly describe bar graphs? Select all that apply.

1 / 1 point

- ☒ The horizontal line of a bar graph, usually placed at bottom, is called the x-axis.

✓ Correct

- ☒ The vertical line of a bar graph usually placed to the left is called the y-axis.

✓ Correct

- ☒ Bar graphs use size contrast to compare two or more values.

✓ Correct

- ☐ Bar graphs use segments to represent the proportions of each data category compared to the whole.